Ginhawa station: complete build steps

Every stage from the battery pack to mounting the station outside, in order. Each stage has: what you need, numbered steps, a **Check**, and what to do if it fails. Do them in order and pass every Check before moving on.

Build order

#	Stage	What you build
1	Battery pack	cells, holder, BMS
2	Power board	+ wire (row 17), − wire (row 02)
3	Fuse + INA219	battery → fuse → INA219 → power board
4	Solar panel + CN3791	panel → charger → power board
5	3.3 V converter	TPS63020
6	5 V converter	MT3608, set to 5.0 V
7	ESP32 board	headers, rails, resistors
8	Arduino IDE + Blink	laptop set-up, first test (can be done any time)
9	SEN55 + INA219	sensor wiring and test
10	4G modem	A7670SA wiring and test
11	Ginhawa app	start it on the laptop
12	Ginhawa program	upload to the station
13	24-hour table test	run on battery for a day
14	Prepare the box	drill holes, mesh, gland, vent
15	Put everything in the box	mount and close
16	Mount outside	pole, panel, app check

Your multimeter (ANENG M118A): quick reminder

You want	Press SELECT until the screen shows	Read it as
Voltage	**V** with the straight-line (DC) symbol	the number is the voltage
Beep (is it connected?)	the **•)))** speaker symbol	**beep = connected**, silence = not connected
Resistance	**Ω**	the number is the resistance

Black probe in **COM**, red in **VΩ** (right socket). Never the 10A socket. **Beep and Ω tests only with the power OFF** (fuse out).

Heat-shrink and wire tags

Where	Tube
22 AWG joints	**2 mm**
18 AWG joints	**2 mm** if it slides over, else **3 mm**
Twisted joints	**4 mm**

Wire	Colour	Tag at both ends
3.3 V (TPS63020 output)	red	**yellow**
5 V (MT3608 output)	red	**blue**
SDA	black	**yellow**
SCL	black	**blue**
Modem TXD / RXD / PWRKEY	black	masking-tape flag "TX" / "RX" / "PWR"
All grounds	black	no tag

Stage 1 – Battery pack

What it does: joins the three 18650 cells into one battery, protected by the **BMS** (cuts the power if the battery is over-charged, drained too low or shorted).

Time: about 1 hour. **Work in pairs:** one reads, one does.

What you need

- **4 Beston 18650 cells** (3 used, 1 spare)
- **3-cell parallel holder** (fulabs)
- **BMS JH-886121, 3.7V 20A** (pads **B+, B−, P+, P−**)
- **18 AWG** red and black, about 50 cm each
- Heat-shrink: **2 mm** (or 3 mm), **4 mm**, **14 mm**
- Masking tape, marker, electrical tape
- Iron, 1 mm solder, flux, toothpick, damp sponge or brass wool, stripper, cutter, multimeter

> ⛔ **Safety:** cells stay in their box until a step tells you to use them. Never let a bare red and black wire touch.

Part A – Match the cells (10 min)

1. Meter on **V**. Measure each cell: red probe on the **+ end** (with the small ring/bump), black on the flat **− end**.

2. Write each voltage on masking tape and stick it on the cell.

3. Only put cells together when they're within **0.05 V** of each other.

Your cells read **3.644, 3.644, 3.778 and 3.785 V**: two matched pairs.

Pair	Cells	Use
High pair	**3.778 + 3.785 V**	put in the holder **now** and build with these two
Low pair	**3.644 + 3.644 V**	raise later with the solar charger (Stage 4), then add one as the 3rd cell

Part B – Test the holder (5 min)

4. Ignore the drawings moulded inside the holder; go by the metal. **All 3 springs must be on the same end.**

5. Meter on **•)))** (no cells inside):

Touch the probes to	Should
Each spring to each other spring	**BEEP**
Each flat contact to each other flat contact	**BEEP**
Any spring to any flat contact	**silence**
Black lead to the springs	**BEEP**
Red lead to the flat contacts	**BEEP**

6. Every cell goes in the **same direction**: **− end on the springs**, **+ end on the flat contacts**.

> ✅ **Check:** put the two high-pair cells in, meter on **V** on the red and black leads: **3.6 – 4.2 V**. About 11 V would mean a series holder (return it). Then **take the cells out again**.

Part C – Extend the holder leads (15 min)

7. Cut **10 cm red** and **10 cm black** 18 AWG.

8. Slide a **4 mm** heat-shrink (2 cm long) onto each 18 AWG wire, 3 cm from the end.

9. Strip **10 mm** off the holder's red lead and the red 18 AWG wire. Twist each one's strands tight.

10. Cross the two bare ends and **twist them around each other along the full 10 mm**, right up to the insulation.

11. Flux on the twist. Heat it from **underneath**, feed solder from the **top** until it **soaks through the whole twist**. Remove the solder, then the iron; hold still 3 seconds.

12. Fold the twist flat along the wire, trim any sharp strand, slide the heat-shrink over and shrink it.

13. Same for **black lead + black 18 AWG**.

> ✅ **Check:** tug each joint firmly: it doesn't move. No bare metal visible.

Part D – Solder to the BMS (20 min)

14. Find the 4 pads on the BMS with your phone camera zoom: **B+, B−, P+, P−**. If the strips are long, cut them to about **5 mm**, then **tin** each pad (flux, iron, a little solder).

15. Cut a **15 cm red** and a **15 cm black** 18 AWG wire. Strip and tin both ends of each.

16. Slide a **2 mm** heat-shrink (or 3 mm if 2 mm won't go over the joint) onto each of the 4 wires, 3 cm from the end.

17. Solder each wire flat onto its pad (press the iron on top 2 – 3 s until both melt together, hold still 3 s):

Wire	BMS pad
Red from the holder	**B+**
Black from the holder	**B−**
New 15 cm red	**P+**
New 15 cm black	**P−**

```
Holder red  ──────── B+          P+ ──────── red 15 cm   → fuse (Stage 3)
                        [ BMS ]
Holder black ──────── B−          P− ──────── black 15 cm → − wire (Stage 3)
```

18. Slide each tube up to its pad and shrink it.

19. Wrap the **free ends** of the two 15 cm wires **separately** in electrical tape.

20. Cover the BMS: **14 mm** heat-shrink (about 5 cm), or wrap it in electrical tape.

> ⛔ **Nothing else ever connects to B−.** All station grounds go to **P−**.

Part E – Final check (5 min)

21. Put the **high-pair cells** in (− on springs, + on flat contacts).

> ✅ **Check:** unwrap the two 15 cm wire ends (keep them apart). Meter on **V**: **P+ to P− = 3.6 – 4.2 V**. Then re-tape them and **take the cells out**.

If something goes wrong

What you see	Why	Fix
0 V at P+/P−, cells read fine	BMS asleep (normal for new ones)	Continue: it wakes when the solar charger is connected (Stage 4)
About 11 V	Series holder	Return it
0 V at the holder too	A cell is backwards or a broken joint	Check cell direction; check joints with •)))
BMS or a wire gets **warm**	Short	**Pull the cells out immediately**; look for bridges between pads
Negative reading	Probes swapped	Swap them; nothing is wrong

Stage 2 – Power board

What it is: a tiny power strip on the **5 × 7 cm** board. Two bare wires on the back: **+ (row 17)** and **− (row 02)**. Every part later gets its power from them.

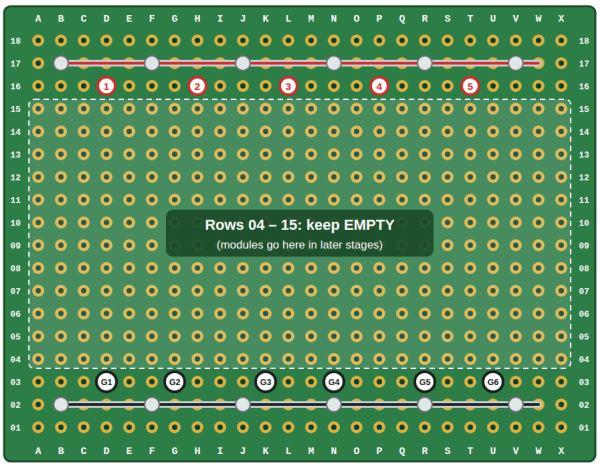
Time: about 40 minutes. No battery needed.

What you need

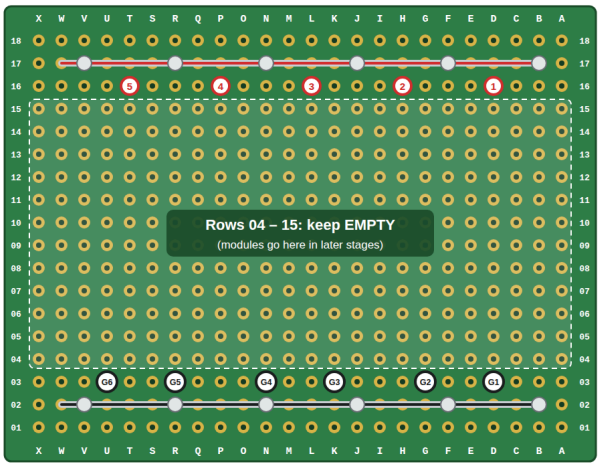
- **5 × 7 cm** perfboard (letters **A – X**, numbers **18 – 01**)
- **18 AWG** red and black, 7 cm each
- Iron, solder, flux, toothpick, stripper, cutter, **2 pliers**, masking tape, marker, multimeter

Ginhawa power board (5 × 7 cm): where the bus wires go

TOP SIDE (printed letters readable, A on the left, 18 at the top)



BOTTOM SIDE (flipped left-to-right: letters are mirrored, X on the left)



VBAT bus: bare red-marked wire on ROW 17, from B17 to W17

GND bus: bare black-marked wire on ROW 02, from B02 to W02

Solder the bus wire here: columns B, F, J, N, R, V (every 4th hole)

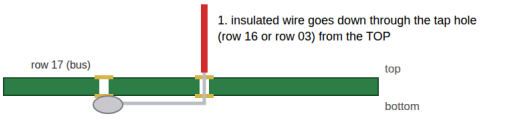
VBAT taps 1–5 (row 16: D16 H16 L16 P16 T16) – used in later stages

GND taps G1–G6 (row 03: D03 G03 K03 N03 R03 U03) – used in later stages

What plugs into each tap later

1	D16	INA219 VIN−	Stage 13	G1	D03	BMS P−	Stage 13
2	H16	CN3791 BAT +	Stage 14	G2	G03	CN3791 BAT −	Stage 14
3	L16	TPS63020 VIN	Stage 15	G3	K03	TPS63020 GND	Stage 15
4	P16	MT3608 VIN+	Stage 16	G4	N03	MT3608 VIN−	Stage 16
5	T16	A7670SA VBAT	Stage 21	G5	R03	A7670SA GND	Stage 21
				G6	U03	ESP32 board GND	Stage 17

How a tap wire joins the bus (later stages)



2. on the bottom, bend the bare end over to the bus wire
3. solder it to the bus (shiny blob)

CHECK with the multimeter on BEEP (no battery connected):

```
B17 ↔ W17 → BEEP (VBAT bus is one wire)
B02 ↔ W02 → BEEP (GND bus is one wire)
row 17 ↔ row 02 → NO beep (no short between + and −)
row 17 ↔ any middle hole (e.g. L10) → NO beep
```

Power board taps: + wire row 17 (taps 1–5 on row 16), − wire row 02 (taps G1–G6 on row 03).

Part A – Know your board (2 min)

1. Front: hold it so the letters read normally, **A on the left**, **18 at the top**. The letters are printed top and bottom, the numbers left and right: same rows and columns, labelled twice.

2. Back: turn it over **like a page in a book**. Row 18 stays at the top; only the letters are mirrored (X on the left). **All soldering in this stage is on the back.**

Part B – Make the two bare wires (10 min)

3. Strip **all** the plastic off the 7 cm red and the 7 cm black wire. Keep a scrap of each colour's plastic.

4. Straighten each with two pliers (pull gently). **Tin** each along its whole length until it's silver.

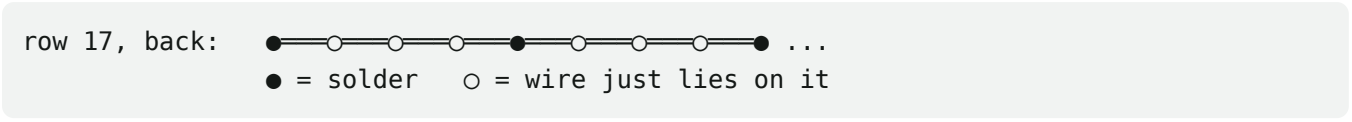
5. Slide the **red scrap** onto one end of the first wire (this is **+**) and the **black scrap** onto the second (this is **−**).

Part C – The + wire on row 17 (10 min)

6. On the **back**, find **row 17**: the **second row from the top**.

7. Lay the + wire **on top of row 17's holes, along their centres**: not in the gap between rows, not pushed into the holes. From the **2nd hole from one edge** to the **2nd hole from the other edge**. Tape both ends.

8. Solder it to the rings at **every 4th hole** (B, F, J, N, R, V). More joints are fine, **as long as the solder stays on row 17**.



9. Remove the tape; press with a fingernail: nothing lifts.

Part D – The − wire on row 02 (10 min)

10. Same on **row 02**: the **second row from the bottom**. Lay, tape, solder every 4th hole, keep the solder on row 02.

Rows 04 – 15 stay empty.

Part E – Mark the front (3 min)

11. Red marker line along **row 17** ("+"), black line along **row 02** ("−").

12. Circle the tap holes for later:

+ taps (row 16)	Later goes to	− taps (row 03)	Later goes to
1 = D16	INA219 VIN− (Stage 3)	**G1 = D03**	BMS P− (Stage 3)
2 = H16	CN3791 BAT + (Stage 4)	**G2 = G03**	CN3791 BAT − (Stage 4)
3 = L16	TPS63020 VIN (Stage 5)	**G3 = K03**	TPS63020 GND (Stage 5)
4 = P16	MT3608 VIN+ (Stage 6)	**G4 = N03**	MT3608 VIN− (Stage 6)
5 = T16	A7670SA VBAT (Stage 10)	**G5 = R03**	A7670SA GND (Stage 10)
		G6 = U03	ESP32 board GND (Stage 7)

Every tap works the same way: the wire goes in from the **front** through its hole; on the **back** you bend the bare end onto the bus wire and solder it.

Part F – Check (5 min)

✅ Check (meter on •))), nothing connected)

Touch the probes to	Should
Both ends of the **+ wire**	**BEEP**
Both ends of the **− wire**	**BEEP**
+ wire and every hole in **row 16** and **row 18** (slide the probe along)	**silence**
− wire and every hole in **row 01** and **row 03**	**silence**
+ wire and the **oval pads** on the edges	**silence**
+ wire and − wire	**silence**

If something goes wrong

What you see	Fix
Beeps to a neighbouring row	Solder bridge: clean tip, flux, drag the iron through the gap, or use the pump. Test again.
A bus doesn't beep end to end	Re-solder the end joints; make sure it's one piece of wire.
Solder won't stick	Flux, clean and re-tin the tip, hold 1 second longer.
A ring lifted off	Leave it; solder to the next hole on the same row.

Stage 3 – Fuse + battery meter (INA219)

What it does: connects the battery pack to the power board. The battery's + goes through the **fuse** (safety and on/off switch) and the **INA219** (measures the battery) to the + wire. The battery's − goes straight to the − wire.

```
BMS P+ ──► fuse holder ──► INA219 VIN+ ─► INA219 VIN− ──► + wire (tap 1 = D16)
BMS P− ──────────────────────────────────────────────────► − wire (tap G1 = D03)
```

Time: about 45 minutes.

What you need

- Battery pack (cells **OUT**) and power board
- **INA219** with its green screw terminal and pin strip
- **Fuse holder** (the one with a wire loop), empty
- **One 5 A blade fuse, the same size as your holder** ("5" printed on top; the number matters, not the colour). 7.5 A is OK if you have no 5 A. Nothing higher.
- **Red 18 AWG**, about 10 cm
- Heat-shrink **2 mm** and **4 mm**
- Iron, solder, flux, stripper, cutter, small screwdriver, multimeter

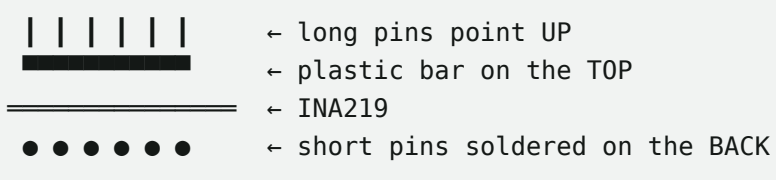
> **Fuse size:** your holder is **standard** size (about 19 mm wide fuse). If your fuses are **mini** (about 11 mm), they won't fit: buy **"standard blade fuse 5 A" (ATC/ATO)**, or buy a **mini** inline holder instead and use your mini fuses.

Part A – Prepare the INA219 (15 min)

1. Find the **2 big holes** (**VIN+**, **VIN−**) and the **6 small holes** (**VCC, GND, SCL, SDA** + 2). The **top** is the side with the chip.

2. Push the **green screw terminal** into the 2 big holes **from the top**, openings facing **outward**. Solder its 2 legs on the back.

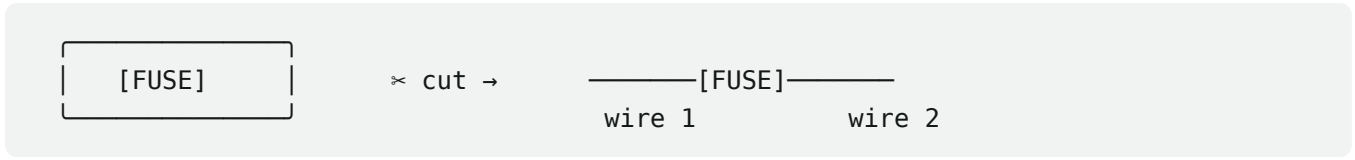
3. Pin strip: from the top. Plastic bar on the chip side, **short pins down through the holes**, long pins pointing up. Solder the 6 short pins on the back (one end pin first, check it's straight, then the rest).



(Or skip the pin strip and solder wires straight into the holes later.)

Part B – Prepare the fuse holder (5 min)

4. No fuse in the holder. Its wire is **one loop**: **cut the loop in the middle**, at the point farthest from the holder. You now have **two wire ends** (identical, either can go either side).



5. Trim each to **10 – 15 cm**. Strip and tin **10 mm** of each.

Part C – Fuse holder to the battery (10 min)

6. Slide a **4 mm** heat-shrink onto the **BMS P+** wire, 3 cm back.

7. Twist P+ with **fuse holder wire 1** (full 10 mm), solder the twist, slide the tube over, shrink.

Part D – INA219 terminal (5 min)

8. Fuse holder wire 2 → VIN+: loosen the screw, push the wire fully in, **tighten firmly**.

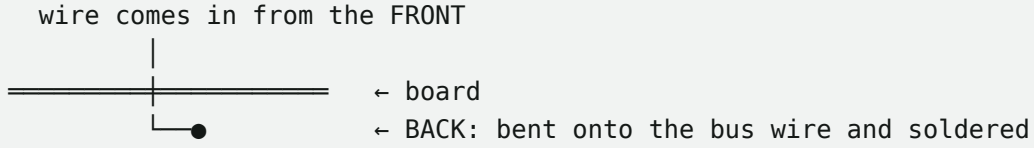
9. 10 cm red 18 AWG: strip 6 mm on one end, 10 mm on the other. **6 mm end → VIN−**, tighten.

> ✅ **Tug test:** pull each wire in the terminal: **it must not move.**

Part E – To the power board (15 min)

10. Red wire from VIN− → tap 1 (D16): in from the **front**; on the **back**, bend it onto the **+ wire** and solder.

11. Black BMS P− wire → tap G1 (D03): strip and tin 10 mm; in from the **front**; on the **back**, bend onto the **− wire** and solder.



12. Trim any bare wire sticking out past the solder.

Part F – Check (5 min)

13. Cells in, **5 A fuse pushed in**.

> ✅ **Check (meter on V): + wire to − wire = 3.6 – 4.2 V**.

14. Pull the fuse out. From now on: **fuse out = everything off.** Always pull it before soldering, and before any •))) or Ω test.

If something goes wrong

What you see	Fix
0 V	Fuse not fully in, a loose terminal, or the BMS asleep (it wakes in Stage 4).
Fuse blows at once	Short: fuse out, cells out, •))) between + and − wire must be silent. Look for bridges.
Wire or INA219 **warm**	Pull the fuse; re-tighten terminals; check for bridges.

Stage 4 – Solar panel + charger (CN3791)

What it does: the panel makes electricity, and the **CN3791** uses it to **charge the battery**, stopping at 4.2 V. It also **wakes the BMS** if it was asleep.

```
Solar panel ──► CN3791 SOLAR ──► CN3791 BAT ──► + wire (tap 2 = H16)
                                              └─► − wire (tap G2 = G03)
```

Time: about 45 minutes.

What you need

- **Solar panel** (junction box "PV-SY017" on the back; cable with **brown** and **blue** wires, ending in a round 2-pin plug)
- **CN3791** (6V version) and its plug cables
- Power board with the battery pack connected (**fuse OUT**)
- Heat-shrink **2 mm**, **4 mm**, **8 mm**; electrical tape; masking tape
- Iron, solder, flux, stripper, cutter, multimeter

Part A – Prepare the panel (15 min)

1. Lay the panel **face down** on a cloth (no electricity while you work).

2. Leave the square **PV-SY017 junction box** alone. If you opened it, close the lid firmly and seal the edge with silicone or electrical tape so rain can't get in.

3. Cut the round plug off at the **far end**, just behind the plug. Keep as much cable as possible: the panel will sit above the box on the pole.

4. Peel back about **5 cm** of the outer jacket. Inside: **brown** and **blue**.

5. Strip and tin **10 mm** of each.

6. Confirm + in direct sun: meter on **V**, panel face up, **red probe on brown, black probe on blue**. A **positive** number means **brown = +**, as expected. Wrap red tape on the **+** wire. Turn the panel face down again.

> ✅ **Check the sun reading**
>
> | Reading | Meaning |
> |---|---|
> | **6 – 7.5 V** | 6 V panel: correct ✅ |
> | **18 – 22 V** | 12 V panel: **stop and tell me** |
> | Below 5 V | Not enough sun; retry at midday |

Part B – Know the CN3791 (2 min)

7. 4 small white sockets: **2 SOLAR** (connected together; use either) and **2 BAT** (connected together; use either).

8. The board shows **+** and **−** next to each socket. Plug a cable in and note which wire colour is on **+**. **Check; don't assume.**

Part C – Panel → SOLAR (10 min)

9. Plug one cable into a **SOLAR** socket, see which wire is **+**, **unplug it**. Strip and tin 10 mm of both wires.

10. Before joining: slide **4 mm** heat-shrink onto each panel wire, plus one **8 mm** piece over both, pushed further back.

11. Twist and solder: brown (+) ↔ cable **+**; **blue (−)** ↔ cable **−**.

12. Shrink the 4 mm pieces, then the 8 mm piece over both. Wrap the joint in **electrical tape**.

Part D – BAT → power board (10 min)

13. Plug a second cable into a **BAT** socket, see which wire is **+**, **unplug it**. Strip and tin 10 mm.

14. Fuse out. + → tap 2 (H16) and **− → tap G2 (G03)**: in from the front, bent onto the bus on the back, soldered.

15. Plug both cables back in: **panel → SOLAR**, **power board → BAT**.

> ⛔ **Double-check:** panel → SOLAR, battery → BAT. Swapped, it can damage the CN3791.

Part E – Test in the sun (10 min)

16. Cells in, **fuse in**, panel **face up in direct sun**.

> ✅ **Check (meter on V)**
>
> | Measure | Expected |
> |---|---|
> | CN3791 SOLAR + and − | **5 – 7.5 V** |
> | Power board + wire and − wire | **3.6 – 4.2 V**, slowly rising, **never above 4.2 V** |
>
> One LED on the CN3791 lights while **charging**, the other when **full**.

17. If Stage 3 read 0 V, it reads the battery voltage now: the charger woke the BMS.

18. Pull the fuse.

Part F – Raise the low-pair cells (when you have sun)

19. Take the high pair out. Put **only the two 3.644 V cells** in.

20. Fuse in, sun, charge **20 – 30 minutes**. Fuse out, cover the panel, **wait 10 minutes**, measure each cell.

21. Repeat until they read **3.76 – 3.80 V**.

22. Final set: **3.778 + 3.785 + one raised cell** (all within 0.05 V). The other raised cell is the spare.

If something goes wrong

What you see	Fix
SOLAR **0 V**	Direct sun; plug fully in; check the panel joint with •))) (fuse out).
No LEDs	A cable's + and − swapped: **fuse out immediately**, recheck each socket.
Voltage doesn't rise	Weak sun or battery already full.
CN3791 **too hot to touch**	Fuse out; check wiring direction. (Warm is normal.)

Stage 5 – 3.3 V converter (TPS63020)

What it does: turns the battery's 3.0 – 4.2 V into a steady **3.3 V** for the ESP32.

Time: 20 minutes.

What you need

- **TPS63020** board (the 3.3 V one)
- **22 AWG** red and black, about 30 cm each
- 2 mm heat-shrink, **yellow** tag pieces
- Foam tape, iron, solder, flux, stripper, multimeter

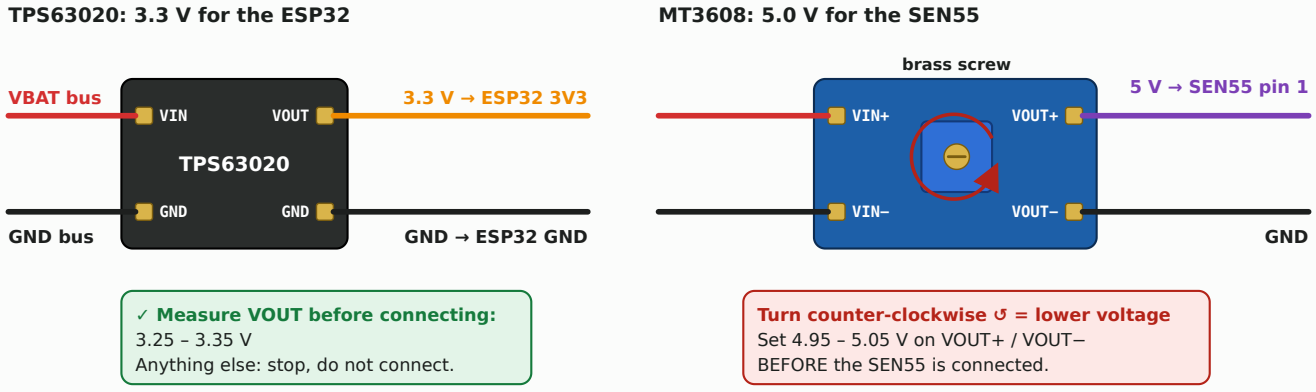


Figure 1. The two voltage converters. Both take power from the VBAT bus. Measure their outputs before anything is connected to them.

Steps

1. Fuse out.

2. Find the 4 pads on the TPS63020 (the names may be on the back): **VIN** and **GND** on the input side, **VOUT** and **GND** on the output side. Take a photo for reference.

3. Stick the TPS63020 on the **front** of the power board, in the empty middle area (rows 04 – 15), with a small piece of **foam tape**.

4. Input wires (10 cm each): - red 22 AWG from **VIN → tap 3 = hole L16** (+ wire); - black 22 AWG from input **GND → tap G3 = hole K03** (− wire).

5. Output wires (20 cm each): - red 22 AWG on **VOUT**, with a **yellow tag** at both ends: this is the **3.3 V wire**; - black 22 AWG on output **GND**.

6. Tape the free ends of the two output wires **separately** so they can't touch.

7. Cells in, **fuse in**. Meter on **V**: measure the yellow-tagged wire (red probe) against its black wire.

✅ **Check**

Measure	Expected
3.3 V wire to its black wire	**3.25 – 3.35 V**

8. Fuse out.

If it fails

What you see	Fix
About **5 V**	It's the 5 V version. **Don't use it for the ESP32.** Contact the seller.
0 V	VIN and GND swapped, or a tap not soldered. Fuse out, check with the beep setting.
Power board reads 0 V too	The BMS may still be asleep: connect the solar charger in sun first.

Stage 6 – 5 V converter (MT3608)

What it does: raises the battery voltage to **5 V** for the SEN55 sensor. **It arrives set much too high**, so you must adjust it before the sensor is ever connected.

Time: 20 minutes.

What you need

- **MT3608** board
- **22 AWG** red and black, about 30 cm each
- 2 mm heat-shrink, **blue** tag pieces
- **Small flat screwdriver**, nail polish or hot glue
- Foam tape, iron, solder, multimeter

Steps

1. Fuse out. Stick the MT3608 on the front of the power board next to the TPS63020 (foam tape).

2. Input wires (10 cm each): - red from **VIN+ → tap 4 = hole P16**; - black from **VIN− → tap G4 = hole N03**.

3. Output wires (20 cm each): - red on **VOUT+**, with a **blue tag** at both ends: this is the **5 V wire**; - black on **VOUT−**.

Tape the free ends separately.

4. Cells in, **fuse in**. Meter on **V**, probes on VOUT+ and VOUT−. It probably reads **6 – 28 V**.

5. Turn the **brass screw** on the blue box **counter-clockwise**, one full turn at a time. **It can take 10 – 25 turns before anything changes.** That's normal.

6. Stop at **5.0 V** (between 4.95 and 5.05). Too low? Turn clockwise a little.

7. Fuse out, fuse in, measure again: still 5.0 V.

8. Put a drop of **nail polish or hot glue** on the brass screw so vibration can't turn it.

9. Fuse out.

✅ **Check**

Measure	Expected
VOUT+ to VOUT−	**4.95 – 5.05 V**

⛔ **Never skip this.** More than 5.5 V destroys the SEN55 instantly.

If it fails

What you see	Fix
Doesn't change after 30 turns	Turn the other way 30 turns; some boards are reversed.
Stuck at about 3.7 V	Input wires swapped or a faulty board.

Stage 7 – ESP32 board

What it is: the ESP32 plugs into sockets on the **7 × 9 cm** board, so it can be removed to program it. The board also has a **3.3 V rail**, a **GND rail** and two small resistors.

Time: 1 – 2 hours.

What you need

- **7 × 9 cm** perfboard (not the practice one if you damaged it)
- **ESP32** (only to set the spacing, then removed)
- **Female header strip**, 2 × **4.7 kΩ** resistors
- 18 AWG black (25 cm), 22 AWG red and black, bare wire for rails
- Iron, solder, flux, cutter, pliers, multimeter

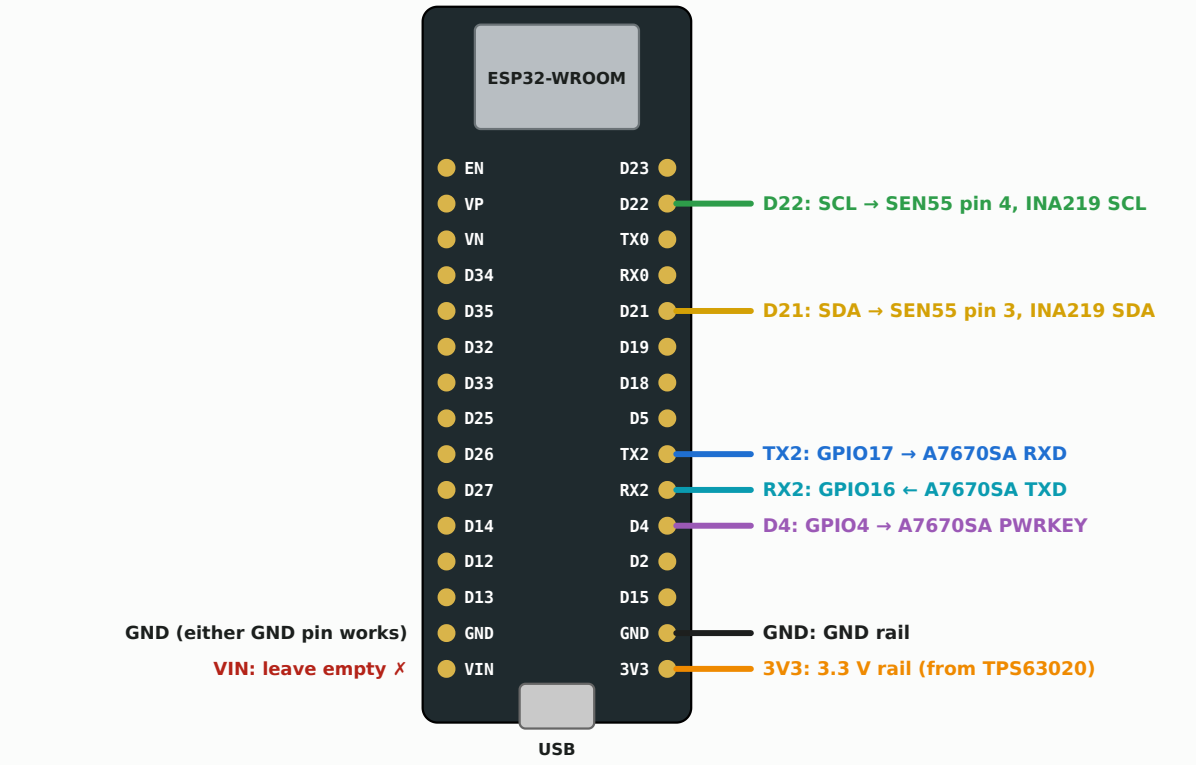


Figure 2. The ESP32 pins this project uses, all on one side of a 30-pin DevKit V1. **TX2 = GPIO17** and **RX2 = GPIO16** on the board's labels.

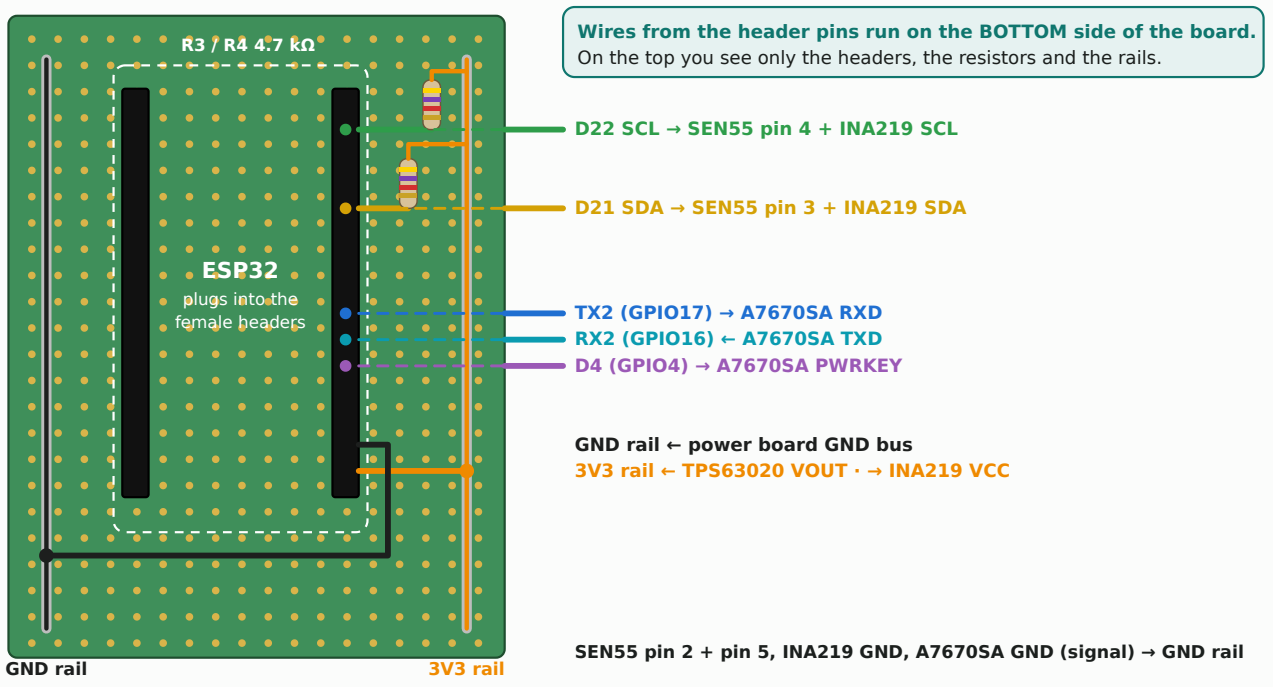


Figure 3. The ESP32 board (7 × 9 cm). Two rails (GND and 3V3), two pull-up resistors, and solder points for the sensor and modem wires.

Part 1 – Headers (40 min)

1. Count the pins on **one side** of your ESP32 (usually **15**).

2. Cut **two** pieces of female header that long: **pull the metal pin out** where you want to cut, then cut through the empty slot and file it smooth.

3. Push both headers onto the ESP32's pins. Set the ESP32 with its headers onto the board, **centred**, USB end at one short edge. This gives the exact spacing.

4. Hold it and turn the board over. **Solder one pin at each end of each header.** Turn it back: check the headers sit **flat and straight**. Then solder **all the other pins**. - With your 1 mm solder: **flux, a tiny dab per pin**, every other pin first.

5. Pull the ESP32 out and put it back in its bag.

> ✅ **Check:** meter on •))), touch each header pin to its neighbour: **silence everywhere** (no bridges).

Part 2 – Rails (20 min)

6. Make two bare, tinned wires (like the power board buses), as long as the board's long edge.

7. On the **back**, solder one along the **left long edge**: the **GND rail**. Solder the other along the **right long edge**: the **3V3 rail**. Mark them on the front with marker ("GND" black, "3V3" red).

8. On the back, with short pieces of wire: - the header pin under the ESP32's **3V3** pin → **3V3 rail**; - the header pin under the ESP32's **GND** pin → **GND rail**.

(Figure above shows where 3V3 and GND are: bottom of the right side, next to each other.)

Part 3 – Pull-up resistors (15 min)

9. Find the header pins under **D21** and **D22**.

10. Solder one **4.7 kΩ resistor** from the **D21** pin to the **3V3 rail**, and another from the **D22** pin to the **3V3 rail**. Resistors have no direction.

Part 4 – Power in (15 min)

11. The **yellow-tagged 3.3 V wire** from the TPS63020 → **3V3 rail**. Its black output wire → **GND rail**.

12. 18 AWG black, 25 cm: one end to the **GND rail**, the other to **tap G6 = hole U03** on the power board.

> ✅ **Check (ESP32 NOT plugged in)**
>
> | Test | Meter | Expected |
> |---|---|---|
> | 3V3 rail to GND rail, **fuse in** | V | **3.25 – 3.35 V** |
> | D21 pin to 3V3 rail, **fuse out** | Ω | about **4.7 kΩ** |
> | VIN pin to anything, **fuse out** | •))) | **silence**: VIN stays empty |

13. Fuse out.

If it fails

What you see	Fix
Headers crooked	Reheat one pin at a time while pressing straight, or remove with the pump and redo.
3V3 rail 0 V	TPS63020 output wire or the U03 ground link not soldered. Check with •))).

Stage 8 – Arduino IDE + first test (Blink)

What it does: installs the free program that uploads code to the ESP32, and proves the laptop, cable and ESP32 work. Do this on the laptop; it doesn't need the rest of the station.

Time: 45 minutes.

What you need

- Laptop with internet
- **ESP32** (on its own, not on the board)
- **USB data cable** (one that can transfer phone files, not charge-only)



Figure 4. Adding the ESP32 boards address in Preferences. You do this once per computer.

Steps

1. Go to **arduino.cc/en/software**. Download **Arduino IDE 2** for Windows (64-bit installer). Install with all defaults; allow drivers if asked.

2. Open it. **File → Preferences**. In **"Additional boards manager URLs"** paste: `https://espressif.github.io/arduino-esp32/package_esp32_index.json` Click **OK**.

3. Click the **board icon** (2nd on the left bar). Search **esp32**. Install **"esp32 by Espressif Systems"**. Wait until it says INSTALLED (large download).

4. Click the **books icon** (3rd on the left). Install these, and click **"Install all"** when asked about dependencies: - **Sensirion I2C SEN5X** - **Adafruit INA219** - **ArduinoJson** (by Benoit Blanchon, version 7)

5. Plug the ESP32 into the laptop. Open **Device Manager** (right-click Start). Under **Ports (COM & LPT)**, a new **COM** number appears (e.g. COM3).

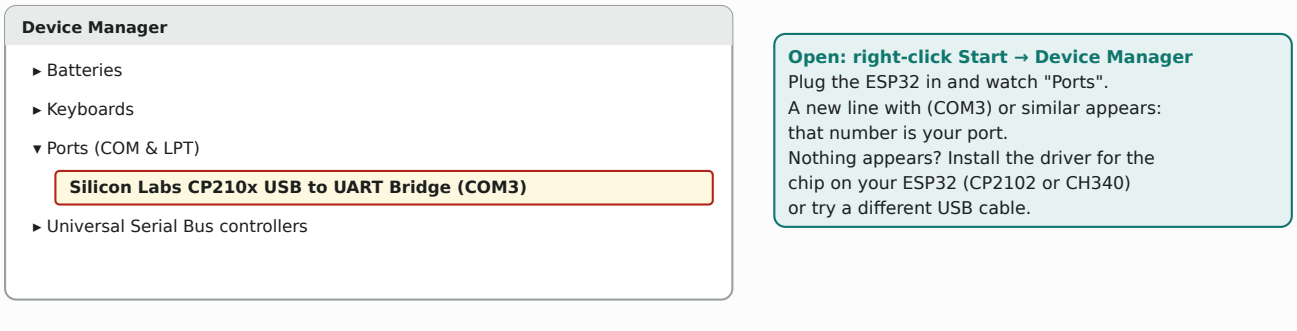


Figure 5. Finding the ESP32's COM port on Windows. The name says CP210x or CH340 depending on the USB chip on your board.

6. No COM port? Try another cable. Then look at the small chip next to the ESP32's USB socket: install the **CP210x** (Silicon Labs) or **CH340** (WCH) driver it names, restart, and plug in again.

7. File → New Sketch. Delete everything, paste the Blink code below.

```
#define LED_BUILTIN 2
void setup() { pinMode(LED_BUILTIN, OUTPUT); }
void loop() {
  digitalWrite(LED_BUILTIN, HIGH); delay(1000);
  digitalWrite(LED_BUILTIN, LOW);  delay(1000);
}
```

8. Top bar → **Select other board and port** → choose **ESP32 Dev Module** and your **COM** port.

9. Click **Upload** (→). If it stays at **"Connecting...."**, **hold the BOOT button** on the ESP32 until percentages appear.

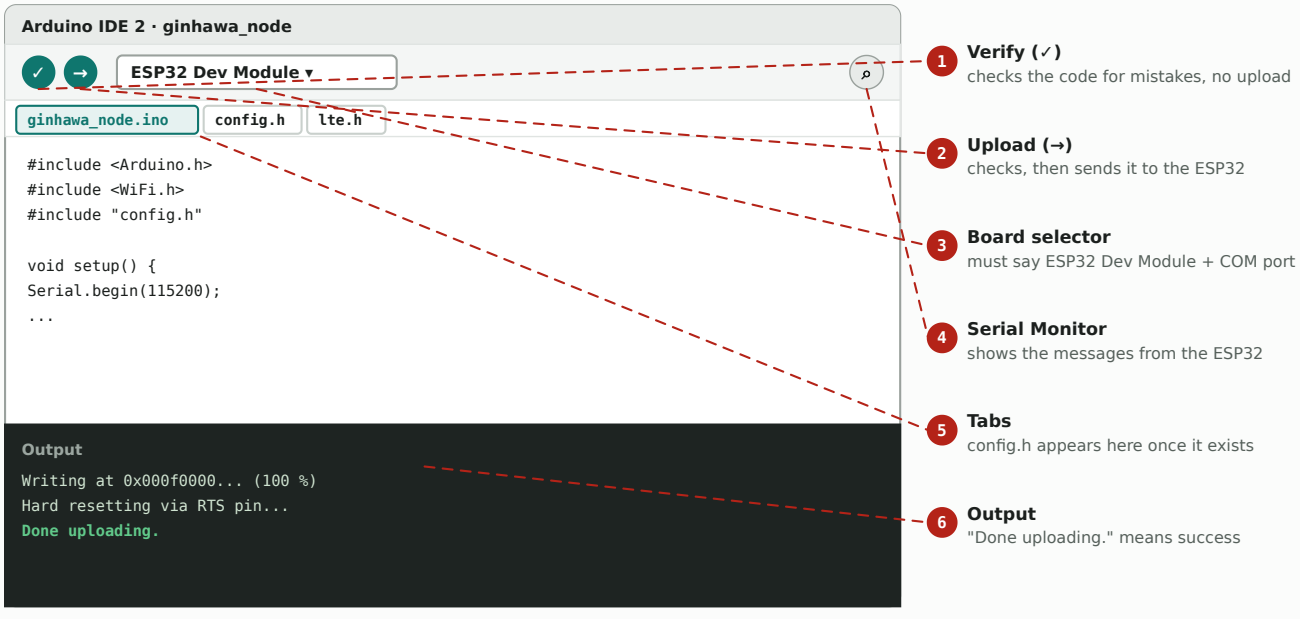


Figure 6. The Arduino IDE 2 window and the six things you will use. (Simplified drawing; colours on your screen may differ.)

✅ **Check:** "Done uploading" and the ESP32's **blue LED blinks** once per second.

If it fails

What you see	Fix
Port greyed out	Another program (e.g. Serial Monitor) is using it; close it.
"Failed to connect"	Hold BOOT during "Connecting"; or Tools → Upload Speed → 115200.
Uploads but no LED	Some boards have no LED on pin 2. "Done uploading" means it worked.

Stage 9 – SEN55 + INA219 wiring and test

What it does: connects the air sensor and the battery meter to the ESP32 using **two shared wires**: **SDA** (D21) and **SCL** (D22).

Time: 1 – 2 hours.

What you need

- **SEN55**, one **JST GH 6-pin cable**
- INA219 (already on the power path)
- 22 AWG black, yellow and blue tag pieces, 2 mm heat-shrink
- Iron, solder, flux, stripper, multimeter, laptop

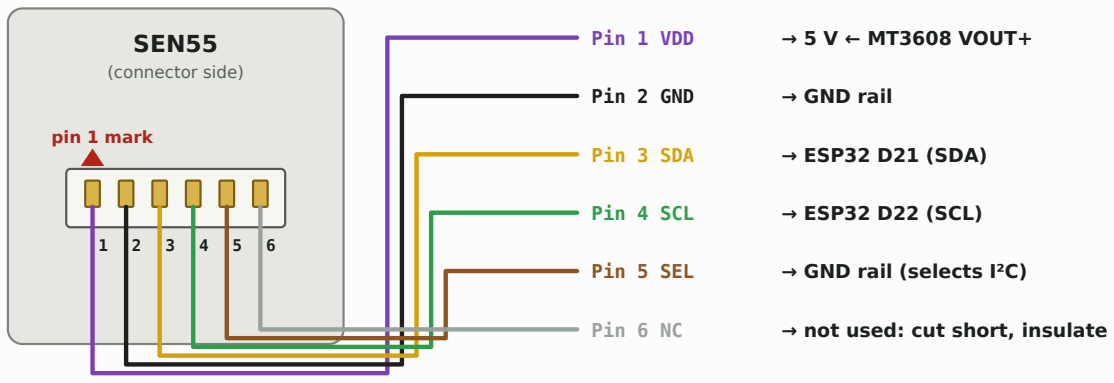


Figure 7. SEN55 pin-out and where each wire goes. Find **pin 1** from the mark printed next to the sensor's connector, then count. Go by pin number, never by wire colour.

Part 1 – Find pin 1 (20 min)

1. Cut **one** double-ended cable exactly **in the middle**. Keep the other half as a spare.

2. Strip **4 mm** of all 6 wires on one half and tin them.

3. Plug it **gently** into the SEN55 (line it up; never force it).

4. Find **pin 1**: a small triangle or "1" next to the SEN55's socket.

5. Meter on •))). One probe on a bare wire end; with the other, touch the metal contact inside the plug at the pin 1 position (a needle helps). Find which wire is pin 1, then 2 – 6.

6. Write it down: pin 1 = (colour), pin 2 = (colour), … **Wire colours on these cables aren't standard: always go by pin number.**

Part 2 – Solder the SEN55 (25 min)

7. Fuse out.

SEN55 pin	Connect to
1 VDD	**blue-tagged 5 V wire** (MT3608 VOUT+)
2 GND	**GND rail** (ESP32 board)
3 SDA	header pin under **D21** (yellow tag)
4 SCL	header pin under **D22** (blue tag)
5 SEL	**GND rail**
6 NC	nothing: cut short, cover with heat-shrink

8. Also solder the MT3608's black **VOUT−** wire to the **GND rail**.

9. Cover every joint with 2 mm heat-shrink.

Part 3 – Solder the INA219 signals (15 min)

10. Four 22 AWG wires from the INA219's pins:

INA219 pin	Connect to
VCC	**3V3 rail**
GND	**GND rail**
SDA	header pin under **D21** (yellow tag)
SCL	header pin under **D22** (blue tag)

Part 4 – Test (20 min)

11. Plug the **ESP32 into its headers**: every pin in a socket, none outside, USB end toward the board edge.

12. Cells in, **fuse in**, then plug the USB cable into the laptop.

13. Arduino IDE: **File → New Sketch**, paste **Test 1** below, **Upload**.

Test 1: I²C scanner

```
// Ginhawa test 1: finds devices on the I2C bus (SDA = D21, SCL = D22)
#include <Wire.h>

void setup() {
  Serial.begin(115200);
  delay(1000);
  Wire.begin();  // SDA 21, SCL 22
}

void loop() {
  Serial.println("Scanning...");
  int found = 0;
  for (byte addr = 1; addr < 127; addr++) {
    Wire.beginTransmission(addr);
    if (Wire.endTransmission() == 0) {
      Serial.printf("  found 0x%02X", addr);
      if (addr == 0x40) Serial.print("  <- INA219 (battery meter)");
      if (addr == 0x69) Serial.print("  <- SEN55 (air sensor)");
      Serial.println();
      found++;
    }
  }
  if (found == 0) Serial.println("  nothing found: check SDA/SCL, power and GND");
  Serial.println();
  delay(3000);
}
```

14. Open the **Serial Monitor** (magnifier, top right). Set it to **115200 baud**.

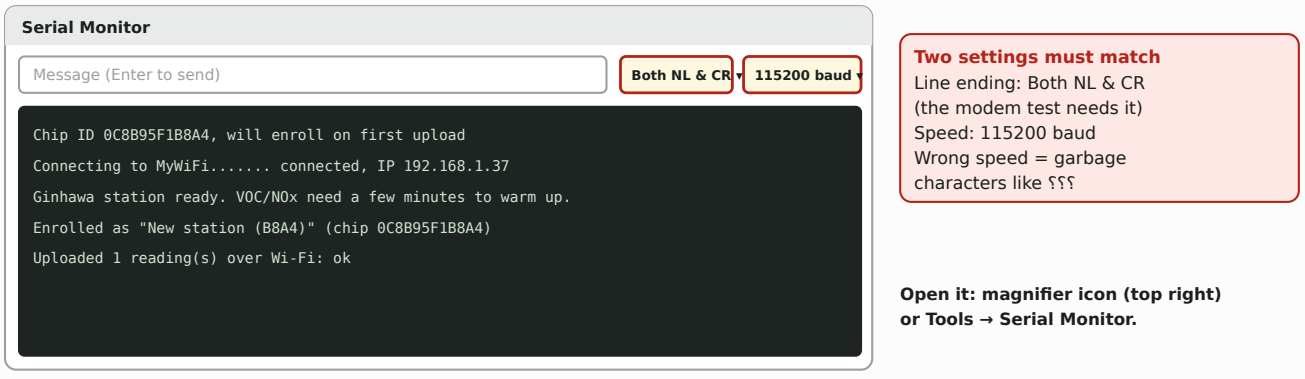


Figure 8. The Serial Monitor shows what the ESP32 is doing. The text is an example of a healthy first start.

✅ **Check:** the Serial Monitor shows, every 3 seconds: `found 0x40 <- INA219 (battery meter)` `found 0x69 <- SEN55 (air sensor)`

If it fails

What you see	Fix
Nothing found	Swap the D21/D22 wires; check the 3V3 rail; check the U03 ground link.
Only 0x40	SEN55 not powered or wrong pin order: measure 5 V between pin 1 and pin 2 wires.
Only 0x69	INA219 VCC or its SDA/SCL wires.
Strange characters	Serial Monitor not at 115200.

Stage 10 – 4G modem (A7670SA) wiring and test

What it does: sends readings over 4G when there's no Wi-Fi, and its GPS finds the station's location.

Time: 1 – 2 hours.

What you need

- **A7670SA** board, **LTE antenna**, **GPS antenna**
- **Nano-SIM** (registered, with data load)
- 2 × **1000 µF** capacitors
- 18 AWG red and black (15 cm each), 22 AWG black, masking tape
- Iron, solder, multimeter, laptop

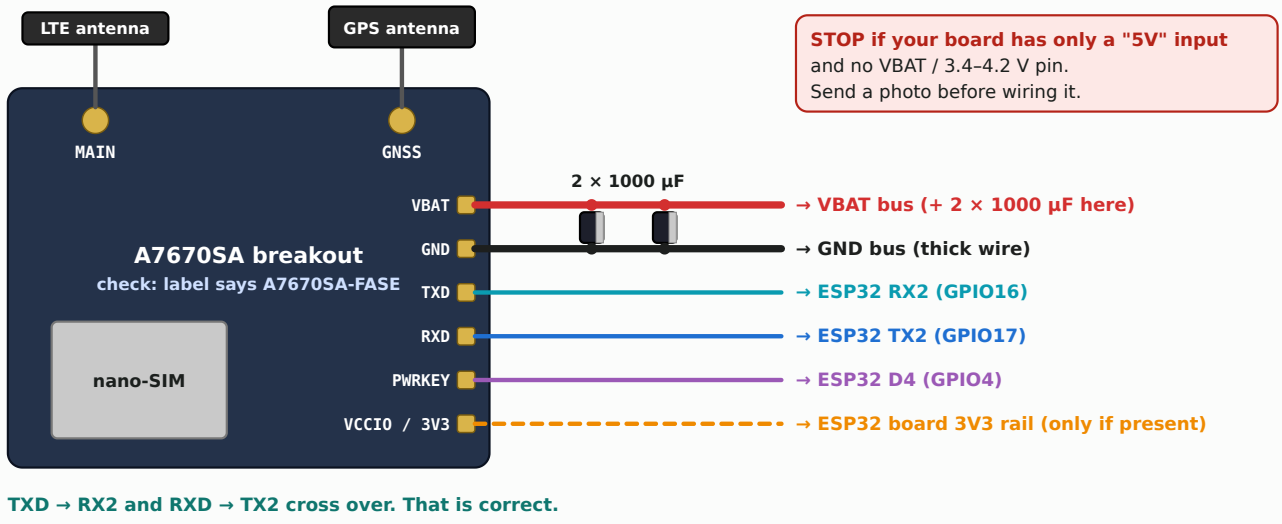


Figure 9. A7670SA connections. Pin names differ between breakout makers: PWRKEY may be printed as **PWR**, **PEN** or **K**; VBAT may be **VCC 3.4–4.2 V** or **BAT**.

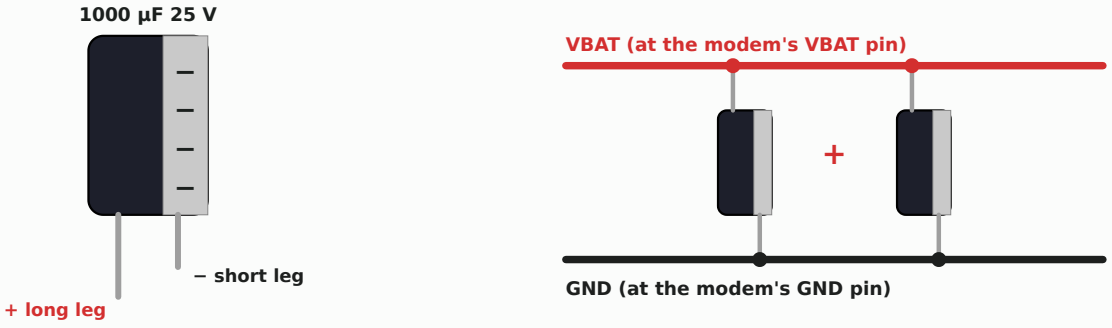


Figure 10. Capacitor polarity. The side with the stripe is **−**. Two capacitors side by side, both + to VBAT and both − to GND.

Steps

1. Find these pins on your board and **write down where they are**: **VBAT** (or VCC 3.4 – 4.2 V / BAT), **GND**, **TXD**, **RXD**, **PWRKEY** (or PWR / PEN / K). Some boards also have **VCCIO** or **3V3**.

2. ⛔ Only a **5V** input and no VBAT pin? **Stop. Send me a photo of both sides first.**

3. Check the shield label says **A7670SA-FASE** (FASE = with GPS).

4. Fuse out. Power wires (keep them **short and straight**): - **red 18 AWG**: power board **tap 5 = hole T16** → modem **VBAT**; - **black 18 AWG**: power board **tap G5 = hole R03** → modem **GND**.

5. Solder the **two 1000 µF capacitors** right at the modem's VBAT and GND: **long leg (+) to VBAT**, **striped side (−) to GND**. Cover the legs with heat-shrink.

6. Signal wires (22 AWG black, about 15 cm, with tape flags): - modem **TXD** → ESP32 board **RX2** pin (GPIO16), flag "TX"; - modem **RXD** → ESP32 board **TX2** pin (GPIO17), flag "RX"; - modem **PWRKEY** → ESP32 board **D4** pin, flag "PWR"; - one thin black wire: modem **GND** → **GND rail**; - if the board has **VCCIO / 3V3**: one wire to the **3V3 rail**.

7. Antennas: LTE antenna → **MAIN** (or LTE); GPS antenna → **GNSS** (or GPS). Press **straight down** until each snaps.

8. Insert the **nano-SIM** (cut corner as drawn on the slot).

9. Fuse in, USB in. Upload **Test 2** below.

Test 2: modem terminal

```
// Ginhawa test 2: talk to the A7670SA. Type AT commands in the Serial Monitor.
// Serial Monitor: 115200 baud, "Both NL & CR".
#define MODEM_RX 16   // ESP32 RX2 <- modem TXD
#define MODEM_TX 17   // ESP32 TX2 -> modem RXD
#define PWRKEY    4   // ESP32 D4  -> modem PWRKEY

void setup() {
  Serial.begin(115200);
  Serial2.begin(115200, SERIAL_8N1, MODEM_RX, MODEM_TX);
  pinMode(PWRKEY, OUTPUT);
  Serial.println("Pressing the modem's power key for 1 second...");
  digitalWrite(PWRKEY, HIGH);
  delay(1000);
  digitalWrite(PWRKEY, LOW);
  Serial.println("Wait about 15 seconds, then type: AT");
}

void loop() {
  while (Serial.available())  Serial2.write(Serial.read());   // you -> modem
  while (Serial2.available()) Serial.write(Serial2.read());   // modem -> you
}
```

10. Serial Monitor: **115200 baud** and **"Both NL & CR"**. Wait **15 seconds**, then type each command and press Enter:

Type	Good answer	Means
`AT`	`OK`	modem is on
`AT+CPIN?`	`+CPIN: READY`	SIM detected
`AT+CSQ`	`+CSQ: 18,99`	signal; first number 10 – 31 is usable
`AT+CEREG?`	`+CEREG: 0,1` or `0,5`	registered on 4G
`AT+CGNSSPWR=1`	`+CGNSSPWR: READY!`	GPS on
`AT+CGPSINFO`	numbers with N and E	position (take it outside, wait 1 – 5 min)

If it fails

What you see	Fix
No answer to `AT`	Swap TXD/RXD wires; measure VBAT at the modem (3.6 – 4.2 V); press the board's own power button if it has one.
`+CPIN: ERROR`	SIM upside down, not registered, or has a PIN (remove it with a phone).
`+CSQ: 99,99`	Antenna on the wrong socket, or no signal indoors: go near a window.
Modem restarts when connecting	Battery low or VBAT wires too long/thin; add the third capacitor.

Stage 11 – Start the Ginhawa app on the laptop

Time: 30 minutes.

1. Make sure **Docker Desktop** is running (whale icon, "Engine running").

2. Open **PowerShell**: `cd C:\Users\peter\ginhawa`

3. Type `ipconfig`. Under **Wireless LAN adapter Wi-Fi**, copy the **IPv4 Address** (e.g. 192.168.1.20).

4. `notepad .env` and set (with your address):

```
SITE_ADDRESS=localhost, 192.168.1.20
DEFAULT_SNI=192.168.1.20
```

Save. Also copy the value after `ENROLL_SECRET=` (you need it next).

5. `docker compose up -d --build` and wait until it finishes.

> ✅ **Check:** open **https://localhost** in the browser, accept the warning (Advanced → Proceed): the Ginhawa app opens.

Stage 12 – Upload the Ginhawa program

Time: 30 minutes.

1. In File Explorer open `C:\Users\peter\ginhawa\firmware\ginhawa_node`.

2. Copy `config.example.h`, paste, rename the copy to **`config.h`** (turn on View → File name extensions so it isn't `config.h.txt`).

3. Double-click `ginhawa_node.ino`. Click the **config.h** tab and change:

```
#define WIFI_SSID      "YourWiFiName"
#define WIFI_PASSWORD  "YourWiFiPassword"
#define API_BASE_URL   "https://192.168.1.20"
#define ENROLL_SECRET  "same-as-ENROLL_SECRET-in-.env"
#define LTE_APN        "internet.dito.ph"
```

- Wi-Fi must be **2.4 GHz** (the ESP32 can't see 5 GHz).
- API_BASE_URL = **https://** + your laptop's address, **no slash at the end**.
- APN: DITO `internet.dito.ph`, Smart `internet`, Globe `internet.globe.com.ph`.

4. Save (Ctrl+S). Select **ESP32 Dev Module** + COM port. **Upload**.

5. Serial Monitor at **115200**.

> ✅ **Check: within 2 minutes you see** 1. `Chip ID … will enroll on first upload` 2. `Connecting to YourWiFiName…... connected` 3. `Ginhawa station ready.` 4. `Enrolled as "New station (…)"` 5. `Uploaded 1 reading(s) over Wi-Fi`
>
> In the app, a new station card says **"Just added"**.

If it fails

Message	Fix
`Connecting … failed`	Wrong name/password, or a 5 GHz network.
`Enrollment failed (HTTP 401/403)`	ENROLL_SECRET doesn't match .env exactly.
`Upload failed (HTTP -1)`	Wrong laptop IP, Docker not running, or different Wi-Fi.
`config.h: No such file`	File is named config.h.txt: rename it.

Stage 13 – 24-hour table test

1. Unplug USB. The station now runs only on its battery.

2. Put the panel in a sunny window or outside; electronics on the table.

3. Check the app every few hours: a reading every minute, PM values change every 5 minutes, battery shown.

4. Breathe near the sensor or light a match nearby (then blow it out): VOC and PM rise, then fall.

5. Touch the converters, modem and fuse holder: **warm is fine, too hot to hold is not.**

6. After 24 hours write down: battery voltage, any gaps in the data. Keep this for the paper.

Stage 14 – Prepare the box

You still need to buy: an **IP65 ABS junction box**, about **150 × 100 × 70 mm** or bigger.

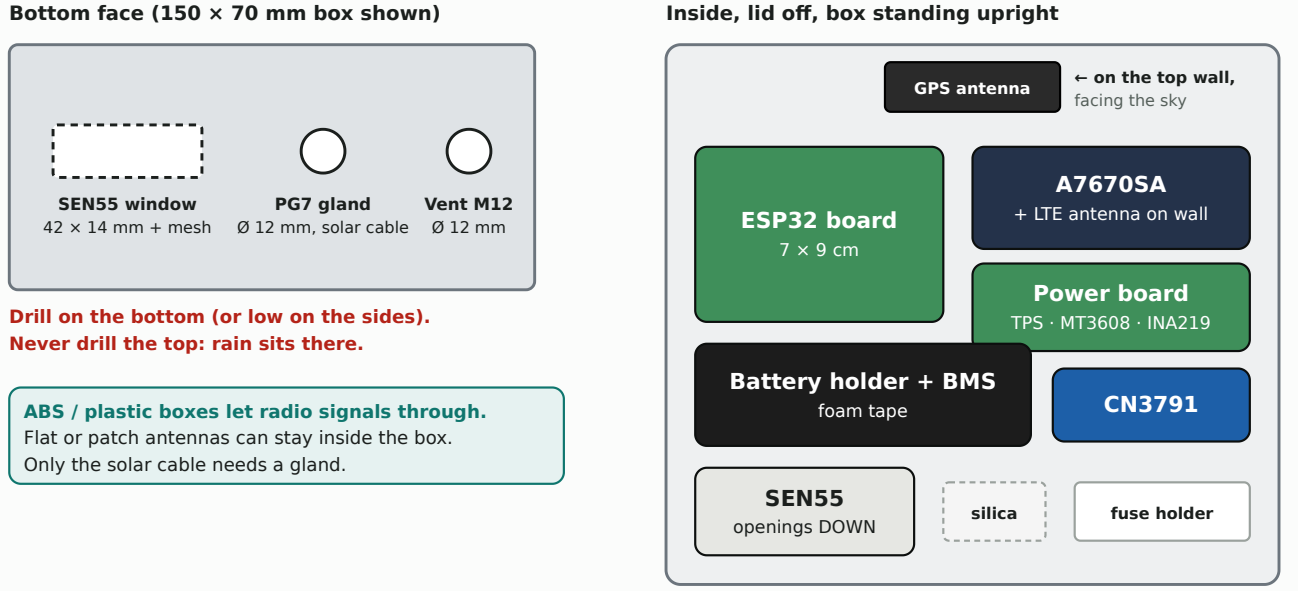


Figure 11. Drilling plan and layout inside an IP65 box of about 150 × 100 × 70 mm. Adjust the positions to the box you buy, keeping these rules.

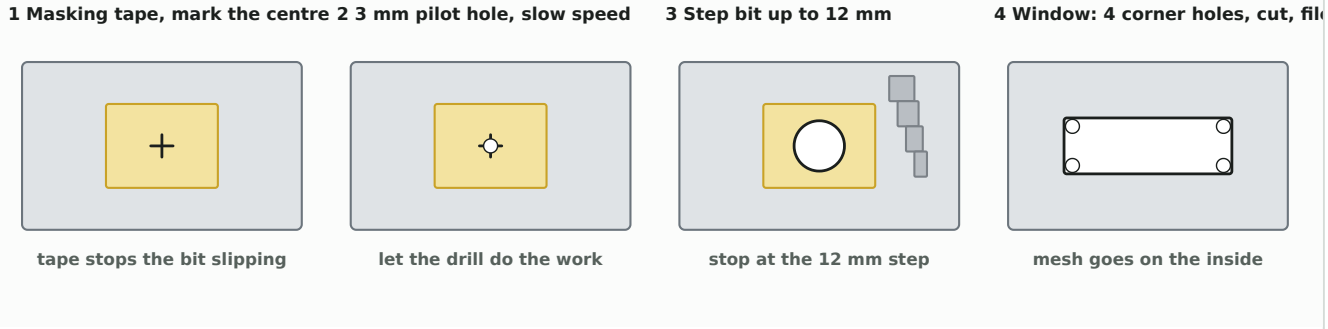


Figure 12. Drilling the plastic box. Go slowly: plastic cracks if you push hard or drill fast.

1. Lay every part inside with the lid on: it must close.

2. On the **bottom face**, put masking tape and mark: **12 mm hole** for the gland, **12 mm hole** for the vent, the **SEN55 window** (42 × 14 mm). Keep 15 mm between holes and edges.

3. Drill **3 mm pilot holes**, slowly, box held firmly.

4. Enlarge to **12 mm** (step bit or 12 mm bit, slow speed).

5. Window: 3 mm holes in the four corners, then cut between them with a cutter (many light cuts). File smooth.

6. Glue the **stainless mesh** on the **inside** of the window, 10 mm bigger on every side, sealed all around.

7. Fit the **PG7 gland** (nut inside) and the **M12 vent** (O-ring outside).

8. Hold the ESP32 board against the back wall, mark its 4 corner holes, drill **3 mm**.

Stage 15 – Put everything in the box

1. Fuse out, cells out.

2. Screw the **4 standoffs** to the back wall (M3 × 8 screws from outside, 2 mm Allen key). Fit the ESP32 board on them, hold with **M3 nuts**.

3. Foam-tape the **battery holder, power board, CN3791 and modem** to the walls. Zip-tie or tape the holder so cells can't fall out.

4. SEN55: openings facing **down**, against the mesh window. At least **3 cm** from the modem and converters.

5. GPS antenna on the inside of the **top** wall, flat side up. **LTE antenna** on a side wall.

6. Solar cable in through the gland from outside, with a **U-shaped drip loop** outside. Tighten the gland.

7. Zip-tie loose wires. **No bare metal touching anything.**

8. Cells in, fuse in. Connect USB briefly: Serial Monitor shows `Uploaded … over Wi-Fi`. Unplug USB.

9. Add 1 – 2 **silica gel** packs. Close the lid evenly (tighten screws a little at a time, in a cross pattern).

> ✅ **Check:** shake gently: nothing rattles. A new reading appears in the app after closing.

Stage 16 – Mount it outside and check the app

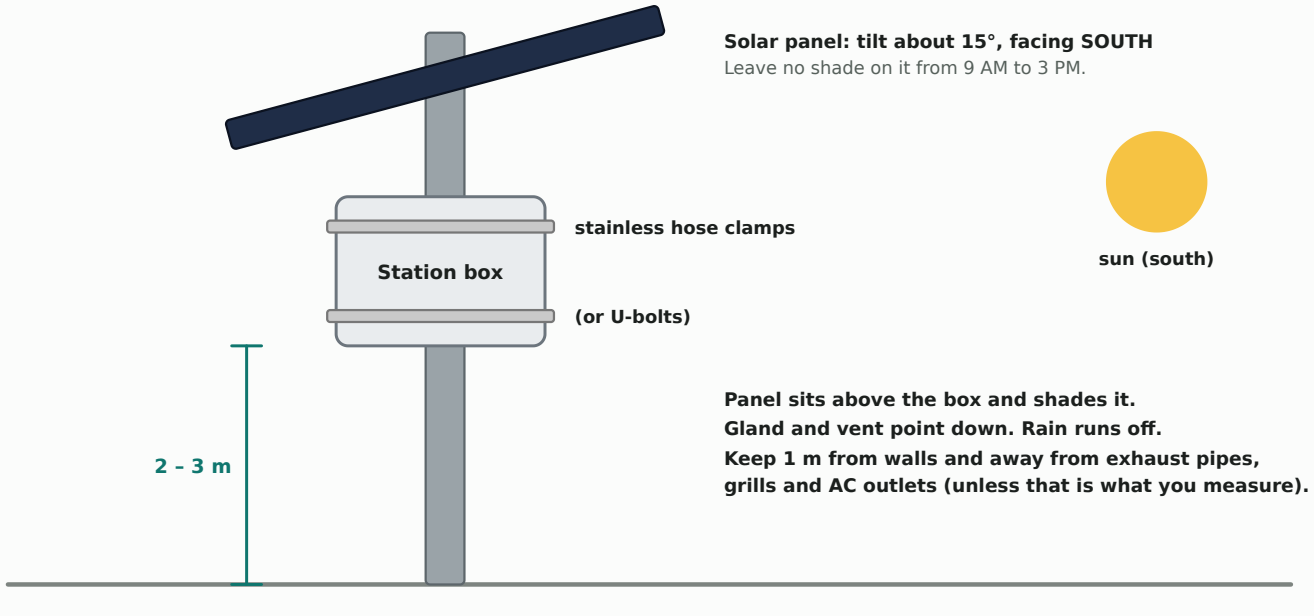


Figure 13. Mounting on a pole. The sensor should breathe the same air people do: 2 to 3 m above the ground, in open air.

1. Choose the spot: open air, **2 – 3 m high**, at least **1 m from walls**, away from exhaust, grills and air-con outlets, with Wi-Fi or mobile signal.

2. Fix the box to the pole with **2 stainless hose clamps** (or U-bolts).

3. Panel **above** the box, tilted about **15°**, facing **south**.

4. Gland and vent **pointing down**. Solar cable zip-tied to the pole with a drip loop.

5. Open the app: readings every minute. Within **15 minutes** outdoors, the GPS finds the position and the station **renames itself** after its barangay.

6. Write down the place, date, chip ID and photos for the paper.

Done. For upkeep (cleaning the panel, replacing silica gel, battery checks) see "Keep it running" in the full manual.